

Smart Revise - System Architecture & Tech Stack

1. TECHNOLOGY STACK USED

Frontend:	HTML5, CSS3, JavaScript (Vanilla)
Backend:	Python (Flask Framework)
Database:	SQLite (SQLAlchemy ORM)
AI / ML:	Large Language Models (Groq / OpenAI API)
Security:	Flask-Bcrypt (Hashing), JWT (Sessions)
Hardware:	Raspberry Pi 4 Model B (IoT Integration)

2. SYSTEM ARCHITECTURE LAYERS

Frontend Layer

Handles User Interaction, Response Rendering, and input collection.

Application Layer

Flask Backend manages business logic, routes, and API calls.

Data Layer

Relational SQLite database for storing persistent user and study data.

Intelligence Layer

AI Engine generates custom revision notes and chatbot replies.

Hardware Layer

Raspberry Pi triggers physical study notifications via GPIO pins.

Security Layer

Ensures encrypted passwords and authenticated user sessions.

3. KEY SYSTEM FEATURES

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- AI-Powered Revision Note Generation
- Real-time AI Study Assistant (Chatbot)
- Dynamic Study Planner and Task Manager
- Daily Login Streaks and Progress Analytics
- Hardware-integrated Study Reminders (Raspberry Pi)